

Role

Act as a Senior Software Architect, Full-Stack Developer, SQA Engineer, DevOps Engineer, System Analyst, and Project Reviewer.

I am providing a ZIP file containing a partially completed Smart University Portal (SUP) project. Your task is to perform a complete technical audit, identify issues, and provide a step-by-step implementation roadmap to transform it into a professional, production-ready academic project.

Phase 1: Project Audit & Analysis

1. Analyze the Entire ZIP Project

Perform a complete codebase review and provide:

A. Project Structure Analysis

- Explain the current architecture.
- Explain frontend structure.
- Explain backend structure.
- Explain database structure.
- Explain API flow.
- Explain module dependencies.

B. Unused Files Detection

Identify:

- Unused files
- Duplicate files
- Dead code
- Unused components
- Unused pages
- Unused APIs
- Unused database tables
- Unused assets

Create a report:

File	Purpose	Used/Unused	Recommendation
Example.js	Login component	Unused	Remove

Provide a final cleanup list so the project becomes professional and maintainable.

Phase 2: Current Functionality Verification

Verify whether the following modules are:

- Fully Working
- Partially Working
- Broken
- Missing

Provide evidence from code.

Authentication Module

Story 1: User Registration

Acceptance Criteria:

- Registration form available
- Email validation
- Password validation
- User record stored

Story 2: User Login

Acceptance Criteria:

- Username validation
- Password validation
- Session creation
- Error handling

Story 3: User Logout

Story 4: Password Reset

Current Issue: Password reset functionality exists but email is not being received.

Analyze:

- Backend implementation

- SMTP configuration
- Email service integration
- Token generation
- Reset workflow

Provide exact fixes.

Student Module Review

Story 5: Student Dashboard

Story 6: Profile Management

Story 7: Course Enrollment

Story 8: Assignment Submission

Acceptance Criteria:

- Upload functionality
- PDF upload
- DOC/DOCX upload
- Submission confirmation
- Timestamp recording
- Database storage

Story 9: Track Assignment Status

Student should see:

- Pending
- Submitted
- Under Review
- Graded

Story 10: View Grades

Student should view:

- Assignment
 - Grade
 - Teacher Feedback
 - Submission Date
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Teacher Module Review

Story 11: Create Assignment

Teacher should be able to:

- Create assignment
- Add title
- Add description
- Add deadline
- Assign course

Story 12: Edit Assignment

Story 13: Review Submission

Teacher should be able to:

- View student submission
- Download PDF/DOCX
- Review work

Story 14: Grade Assignment

Teacher should be able to:

- Enter marks
- Enter feedback
- Save evaluation

Story 15: Manage Courses

Critical Integration Problem

The biggest issue is the integration between Student and Teacher modules.

Required Workflow:

Teacher Creates Assignment ↓ Student Sees Assignment ↓ Student Uploads PDF/DOC/DOCX ↓ Submission Saved ↓ Teacher Views Submission ↓ Teacher Reviews Submission ↓ Teacher Grades Submission ↓ Grade Stored ↓ Student Views Grade & Feedback

Analyze:

- Database relationships
- API endpoints
- Controllers
- Services
- Frontend pages
- Role-based access

Identify exactly where the workflow breaks and provide fixes.

Phase 3: UI/UX Review

Current Problems:

- Poor UI
- Poor UX
- Inconsistent styling
- Unprofessional appearance

Perform a UI audit.

Provide:

Missing UI Components

Design Problems

User Experience Problems

Navigation Problems

Responsive Design Problems

Recommend:

- Modern dashboard design
- Better color palette
- Better typography
- Better forms
- Better tables
- Better assignment workflow UI
- Better admin dashboard UI

Provide implementation steps.

Phase 4: Error Analysis

Identify:

Unit-Level Issues

- Component errors
- Function errors
- Validation issues

Integration Issues

- API failures
- Module communication failures
- Database relationship issues

System-Level Issues

- Authentication issues
- Authorization issues
- Security issues

Provide fixes with implementation order.

Phase 5: Remaining Development Requirements

Implement and design the following missing modules.

ADMIN MODULE

Story 16: Manage Users

Admin can:

- View users
- Add users
- Edit users
- Delete users

Story 17: Assign Roles

Roles:

- Student
- Teacher
- Admin

Story 18: System Monitoring

Admin can monitor:

- User activity
- Login history
- Assignment statistics

Story 19: Generate Reports

Generate:

- Student reports
- Teacher reports
- Course reports
- Assignment reports

Export:

- PDF
- Excel

Story 20: Manage Academic Records

Admin can:

- Manage courses
- Manage enrollments
- Manage academic data

Phase 6: Testing Requirements

Create complete testing strategy.

Story 21: Functional Testing

Provide:

- Test cases
 - Expected results
 - Test data
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Story 22: Automation Testing

Implement Selenium automation for:

- Registration
- Login
- Assignment submission
- Assignment grading

Provide:

- Selenium scripts
 - Folder structure
 - Execution instructions
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Story 23: Regression Testing

Provide:

- Regression test suite
 - Regression checklist
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Story 24: Bug Tracking

Provide:

- Bug report template
 - Severity matrix
 - Priority matrix
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Phase 7: Deployment & DevOps

Story 25: CI/CD Pipeline

Create:

- GitHub Actions workflow
 - Automated build
 - Automated testing
 - Automated deployment
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Story 26: AWS Deployment

Provide deployment architecture:

- Frontend hosting
- Backend hosting
- Database hosting
- File storage

Recommend AWS services and implementation steps.

Story 27: Monitoring Setup

Implement:

- CloudWatch
 - Logging
 - Error monitoring
 - Performance monitoring
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Story 28: Backup & Recovery

Implement:

- Database backup
 - File backup
 - Recovery strategy
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Phase 8: Deliverables

After analyzing the project, provide:

1. Current Status Report

- Completed Features
 - Broken Features
 - Missing Features
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2. Gap Analysis

Compare current project against all 28 Jira stories.

3. Database Review

Provide:

- ERD
 - Missing tables
 - Missing relationships
 - Recommended schema improvements
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4. API Review

Provide:

- Existing APIs
 - Missing APIs
 - Incorrect APIs
 - Required new APIs
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5. Development Roadmap

Provide implementation order:

Priority 1 (Critical)

Priority 2 (High)

Priority 3 (Medium)

Priority 4 (Low)

6. Project Completion Plan

Create a complete roadmap from current state to final production-ready submission.

Include:

- Tasks
 - Dependencies
 - Estimated effort
 - Risks
 - Recommendations
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Expected Output Format

Provide the response in the following order:

1. Project Structure Analysis
2. Unused Files Report
3. Current Functionality Audit
4. Authentication Issues & Fixes
5. Student Module Review
6. Teacher Module Review
7. Student-Teacher Integration Analysis
8. UI/UX Improvement Plan
9. Admin Module Design
10. Testing Strategy
11. Selenium Automation Plan
12. Database Improvements
13. API Improvements
14. AWS Deployment Architecture
15. CI/CD Pipeline Design
16. Monitoring & Backup Strategy
17. Gap Analysis Against All 28 Stories
18. Step-by-Step Development Roadmap
19. Final Professional Recommendations

Do not provide generic suggestions. Analyze the actual codebase and provide implementation-level solutions with file-level references wherever possible.